0613-403 · Week 1 · Lecture 1

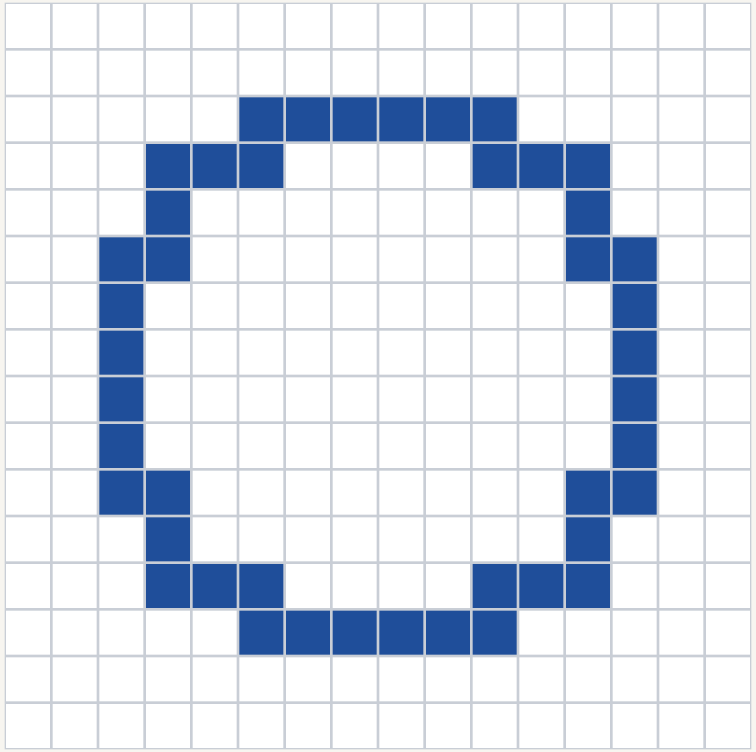
Computer Graphics and Multimedia

Lecture 1: Introduction to Computer Graphics

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Course at a glance

Theory · 0613-403

Credit	3
Contact hours	3 per week
Total marks	100
Classes	2 per week

Lab · 0613-404

Credit	1
Contact hours	2 per week
Total marks	50
Labs	1 per week

Aim: learn the theory, technology, procedures and skills of computer graphics and multimedia, and build 2D and 3D graphics yourself.

Theory outcomes: what you will be able to do

CLO	After this course you can…	Level
CLO1	Explain the basics of computer graphics, its applications, the maths behind it and the basics of multimedia	Understand
CLO2	Identify a suitable drawing algorithm for a given problem	Analyze
CLO3	Illustrate how existing algorithms work and how they are developed	Analyze

Checked through class Q/A, assignments and demonstrations.

Lab outcomes

CLO	After the lab course you can…	Checked by
CLO1	Explain common ideas and techniques of computer graphics	Lab work, lab report
CLO2	Write programs that design and draw graphics scenes	Lab report, lab work
CLO3	Design and improve existing graphics algorithms	Lab work, project
CLO4	Build a project that uses clipping and transformation	Project report, Q/A

The course in 14 weeks

Weeks 1–5

2D graphics

- Pixels and displays
- Lines and circles
- Filling shapes
- 2D transformations
- Clipping

Weeks 6–10

3D graphics

- 3D transformations
- Projection
- Curves
- Hidden surfaces
- Light and shading

Weeks 11–13

Multimedia

- Audio
- Compression
- Video
- Animation
- Virtual reality

Week 14

Projects

- Presentations
- Final review
- Project demo and viva

Lab tools: C++ with OpenGL, and WebGL

C++ with OpenGL (GLUT)

A standard graphics library used in industry and textbooks.

Used for: lines, circles, filling, 2D transformations, clipping, curves, depth buffer, animation.

WebGL

Runs 3D graphics in the web browser using the GPU.

Used for: 3D transformations, lighting and shading.

Each lab ends with one small, visible result. Submit your code and a screenshot of the output.

Lecture 1

What is Computer Graphics?

Definition · history · applications · pixels and resolution · raster vs vector · the graphics pipeline

Computer graphics

The branch of computer science that deals with generating images with the help of computers.

Create

Draw shapes from maths

Store

Save them as pixels or shapes

Change

Move, rotate, resize, recolour

Display

Show on a screen or printer

A short history of computer graphics

1963

Sketchpad

Ivan Sutherland's MIT PhD project. The first interactive drawing program, used with a light pen. The start of CAD.

1970s–80s

CPU drawing

The CPU did all the work. Only 1–10 frames per second, mostly wireframe pictures.

1990s

Graphics cards

Cards such as 3dfx Voodoo and NVIDIA Riva did rasterization in hardware, with fixed steps.

2000s

Shaders

Programmers could write small vertex and fragment programs (DirectX 8, OpenGL).

2010s–now

GPU computing

GPUs also do general computing (CUDA, OpenCL) and even real-time ray tracing.

Graphics has won the top prize in computing twice

Turing Award 1988

Ivan Sutherland

For Sketchpad and his pioneering work that started computer graphics.

Turing Award 2019

Ed Catmull and Pat Hanrahan

For 3D graphics methods behind computer-animated films such as those made by Pixar.

New graphics research is presented at ACM SIGGRAPH, SIGGRAPH Asia and Eurographics.

Where computer graphics is used

Computer art

Cartoons, paintings, logo design

Computer-aided design

Buildings, cars, aircraft

Presentation graphics

Charts and reports for business and science

Entertainment

Films, TV, music videos, games

Education

Simulations and e-learning that are easy to understand

User interfaces

Icons, windows, menus, buttons

How a picture reaches the screen

Input

User gives commands with a keyboard, mouse or tablet

→

Processing

The program and the GPU turn shapes into pixels

→

Frame buffer

Memory that stores the colour of every pixel

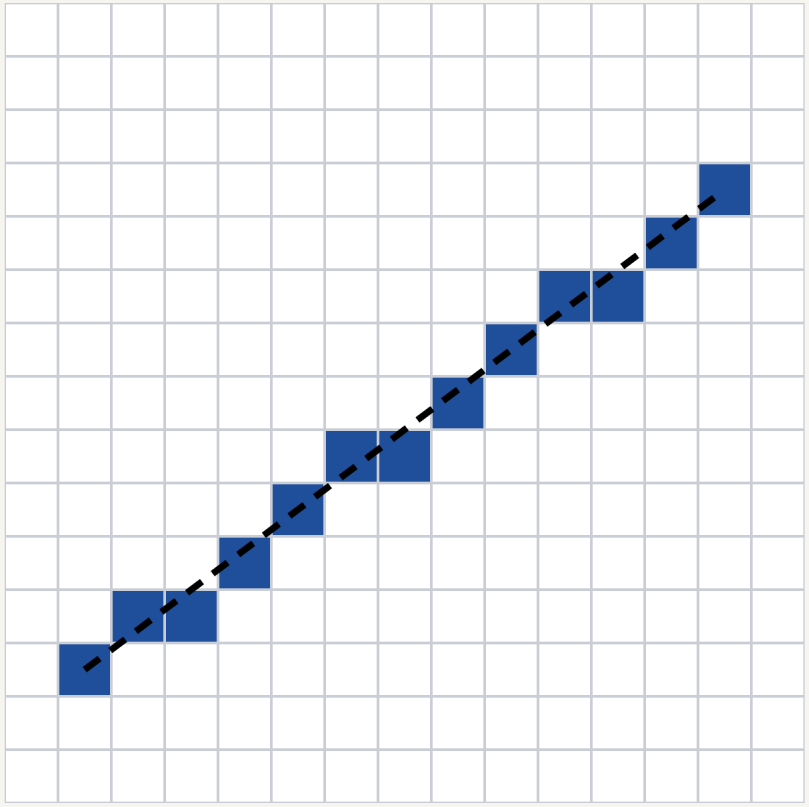
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Display

The monitor reads the frame buffer and redraws many times a second

Interaction: the user sees the picture, responds with new input, and the loop starts again.

Pixel: the smallest dot on the screen



Pixel = picture element

- A screen is a grid of rows and columns of pixels
- Each pixel shows one colour
- A pixel's position is given by (x, y)
- A straight line (dashed) must be drawn using whole pixels (blue)

Resolution: how many pixels

Resolution = number of pixels across × number of pixels down

Name	Resolution	Total pixels
HD	1280 × 720	921,600
Full HD	1920 × 1080	2,073,600
4K	3840 × 2160	8,294,400

Worked example

Memory for one Full HD screen in true colour (24 bits = 3 bytes per pixel):

1920 × 1080 × 3 = 6,220,800 bytes

≈ 5.9 MB for a single frame

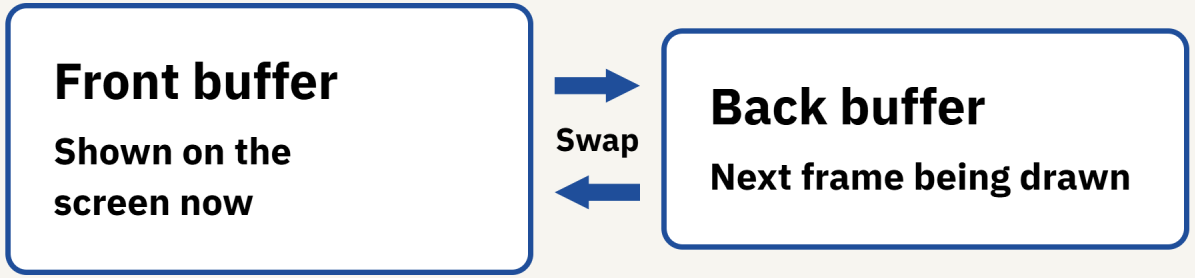
This is why images and video are compressed.

Frame buffer and double buffering

The frame buffer is the memory that holds the colour of every pixel on the screen.

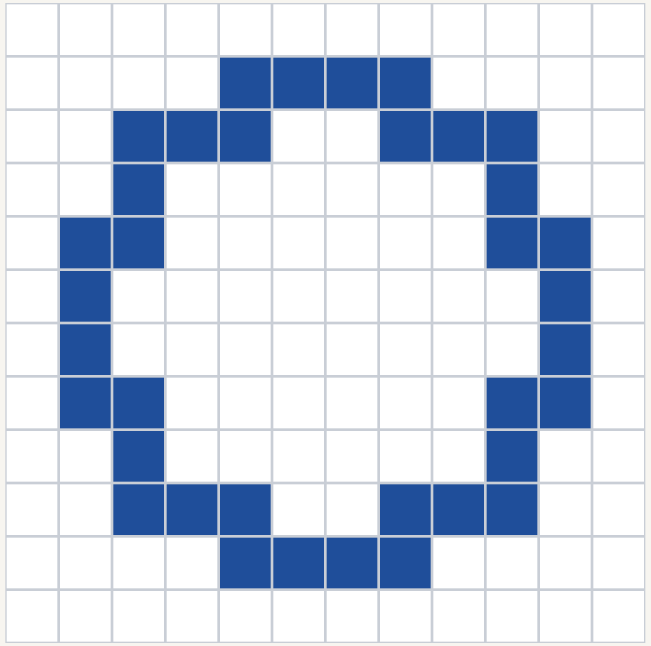
Its size = resolution × bytes per pixel. For Full HD that is about 5.9 MB.

Problem: if we draw straight into it, viewers see half-drawn pictures and flicker.



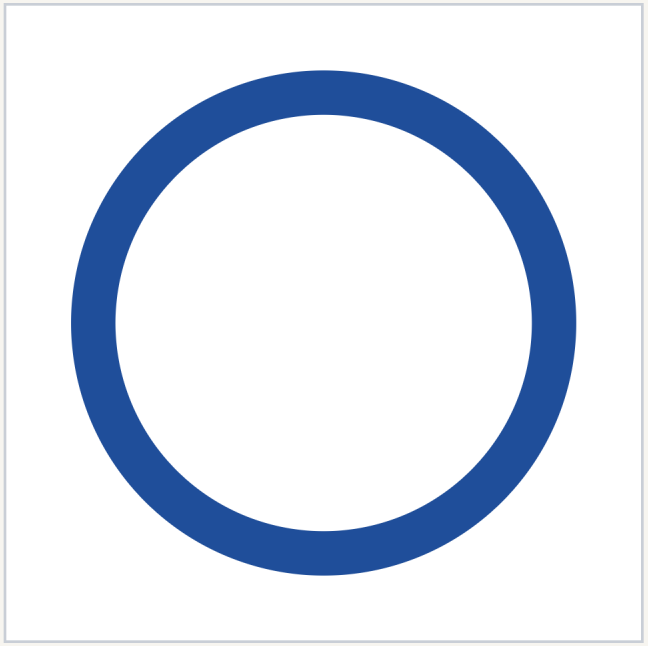
1. Draw the next frame in the back buffer.
2. When it is finished, swap the two buffers.
3. The screen always shows a complete frame: no flicker.

Two ways to store a picture



Raster

Made of pixels. Looks blocky when enlarged.



Vector

Made of lines and curves defined by maths. Stays sharp at any size.

Raster vs vector at a glance

	Raster	Vector
Made of	Pixels	Lines, curves and shapes (maths)
When enlarged	Becomes blocky	Stays sharp
Best for	Photos and detailed images	Logos, diagrams, fonts
File size depends on	Resolution	Number of shapes
Formats	JPEG, PNG, GIF, BMP	SVG, EPS, AI

Graphics, image processing and visualization

Computer graphics

Description → picture

Builds an image from shapes and maths

Image processing

Picture → better picture

Sharpens, removes noise, compresses, enhances

Visualization

Data → picture

Turns numbers into charts, weather maps, MRI scans

Why graphics runs on the GPU

	CPU	GPU
Cores	A few powerful cores (about 4–16)	Thousands of simple cores
Good at	Step-by-step tasks with complex decisions	The same simple job on lots of data at once
Aim	One answer quickly	Many answers every second

Graphics suits the GPU: each vertex and each pixel can be worked out on its own, all at the same time.

The graphics pipeline: from 3D model to pixels

1 Model and view

Place objects and the camera using transformations

Weeks 4 and 6

2 Project

Flatten the 3D scene onto the 2D screen

Week 6

3 Clip

Cut away everything outside the view

Week 5

4 Rasterize

Turn each shape into pixels (scan conversion)

Weeks 2 and 3

5 Hide and shade

Keep the nearest surface; add light, colour and texture

Weeks 9 and 10

6 Frame buffer

Store the finished pixels and show them

Week 1

Real time means fast: the frame budget

60 frames per second

1 ÷ 60 = 16.67 ms per frame

Everything for one frame must finish inside this time.

Frame rate	Time per frame
30 FPS	33.3 ms
60 FPS	16.7 ms
120 FPS	8.3 ms

Example budget for one frame of a game at 60 FPS:



Input 1 ms **Game logic 1.5 ms** **Graphics 12 ms** **Show on screen 0.5 ms** **Spare ≈ 1.7 ms**

Two ways to render a 3D scene

Rasterization

Project each triangle onto the screen, then fill in its pixels.

- Very fast: 60–240 FPS on a GPU
- Used in games and interactive apps
- The main method in this course

Ray tracing

Shoot a ray from the eye through each pixel and find what it hits.

- Realistic shadows and reflections
- Much slower; long used for films
- Now possible in real time on newer GPUs

The choice is a trade-off: speed or physical accuracy.

Where multimedia fits

Multimedia is the computer-controlled combination of different media, all stored digitally:

Text	**Graphics**	**Images**	**Audio**	**Video**	**Animation**

Graphics is one part of multimedia. We study audio, video, compression and animation in Weeks 11–13.

Key points from today

1. Computer graphics means generating images with computers.
2. It is used in art, design, charts, entertainment, education and user interfaces.
3. A picture travels: input → processing → frame buffer → display.
4. A screen is a grid of pixels; resolution is how many pixels it has.
5. Raster images are made of pixels; vector images are made of shapes.
6. GPUs draw fast by running a pipeline of stages in parallel.

Practice questions

1. Define computer graphics and give four applications.
2. What is a pixel? What does resolution mean?
3. How many bytes does an uncompressed 1280 × 720 true-colour (24-bit) image need?
4. Compare raster and vector graphics. Which is better for a company logo, and why?
5. What is the role of the frame buffer? Why do we use double buffering?
6. At 60 frames per second, how much time does the computer have to draw one frame?

Coming up

Next class

Colour and display devices

RGB colour, lookup tables, raster scan displays, printers, plotters, digitizers, light pens

This week's lab

Lab 1: Getting started with OpenGL

Set up C++ with GLUT, open a window, draw points, lines and a coordinate grid

Before the lab: install a C++ compiler with FreeGLUT (for example Code::Blocks or Visual Studio).

Questions?

Thank you

Ekramul Haque Amin · Lecturer, AI and Data Science